

Homework 03

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Question 1: Build an `explain-to-me` skill

I created and installed an `explain-to-me` skill and explicitly used it to explain Question 2. It was designed to clarify the goal, notation, required tasks, and relevant concepts without giving calculations, code solutions, or final answers. The complete skill is included at the end of this report.

Question 2: Ridge regression with nearly collinear predictors

The first two predictors have correlation 0.9983. The largest and smallest singular values are approximately 14.1730 and 0.4061. Because $X_1 \approx X_2$, the common direction is well identified but the difference direction is weakly identified. Thus $\hat{\beta}_1 - \hat{\beta}_2$ should be much less stable than $\hat{\beta}_1 + \hat{\beta}_2$.

λ	$\hat{\beta}_1$ mean (sd)	$\hat{\beta}_2$ mean (sd)	sum mean (sd)	difference mean (sd)	train MSE	test MSE
0	1.290 (1.795)	1.701 (1.798)	2.992 (0.105)	-0.411 (3.592)	0.949	1.038
0.02	1.465 (0.145)	1.497 (0.148)	2.961 (0.104)	-0.032 (0.274)	0.960	1.029
0.2	1.355 (0.049)	1.359 (0.050)	2.715 (0.095)	-0.004 (0.029)	1.064	1.125

OLS has little bias on average but very large variability in the separate coefficients. Ridge greatly reduces that variance. At $\lambda = 0.02$, the variance reduction slightly improves test MSE; at $\lambda = 0.2$, stronger shrinkage creates enough bias to increase test MSE. Predictions change less than the individual coefficients because, when $X_1 \approx X_2$, fitted values depend mainly on $\beta_1 + \beta_2$.

Question 3: What does ridge regression shrink?

For

$$L_\lambda(\beta) = \frac{1}{2n} \|\tilde{\mathbf{y}} - \mathbf{X}\beta\|_2^2 + \frac{\lambda}{2} \|\beta\|_2^2,$$

$$\nabla L_\lambda(\beta) = -\frac{1}{n} \mathbf{X}^\top (\tilde{\mathbf{y}} - \mathbf{X}\beta) + \lambda\beta.$$

Setting the gradient to zero gives

$$(\mathbf{X}^\top \mathbf{X} + n\lambda \mathbf{I}_p) \hat{\beta}_\lambda = \mathbf{X}^\top \tilde{\mathbf{y}}.$$

For any nonzero $\mathbf{a}$,

$$\mathbf{a}^\top (\mathbf{X}^\top \mathbf{X} + n\lambda \mathbf{I}_p) \mathbf{a} = \|\mathbf{X}\mathbf{a}\|_2^2 + n\lambda \|\mathbf{a}\|_2^2 > 0$$

when $\lambda > 0$. Hence the matrix is positive definite and invertible, so the ridge solution is unique even if X is rank deficient or $p > n$.

With compact SVD $\mathbf{X} = \mathbf{U}_r \mathbf{D}_r \mathbf{V}_r^\top$,

$$\mathbf{X} \hat{\beta}_\lambda = \sum_{j=1}^r \frac{d_j^2}{\underbrace{d_j^2 + n\lambda}_{\rho_j(\lambda)}} \mathbf{u}_j \mathbf{u}_j^\top \tilde{\mathbf{y}}.$$

For $n = 100, \lambda = 0.01, n\lambda = 1$, so

$$\rho_1 = \frac{100}{101} = 0.9901, \quad \rho_2 = \frac{4}{5} = 0.8, \quad \rho_3 = \frac{0.04}{1.04} = 0.0385.$$

Including the unpenalized intercept,

$$df_{\text{eff}} = 1 + \rho_1 + \rho_2 + \rho_3 \approx 2.8286.$$

The third singular direction is shrunk most because d_3 is smallest. This lowers variance in a weakly identified direction, but if the true signal is important there, strong shrinkage creates bias.

Question 4: Ridge regression and optimization

The unpenalized condition number is

$$\kappa(\mathbf{A}_0) = \frac{9}{0.01} = 900.$$

With $\lambda = 0.09$, the eigenvalues become 9.09 and 0.10, giving

$$\kappa(\mathbf{A}_\lambda) = \frac{9.09}{0.10} = 90.9.$$

Thus ridge improves conditioning.

Let $\mathbf{e}^{(k)} = \beta^{(k)} - \hat{\beta}_\lambda$. For the quadratic objective,

$$\mathbf{e}^{(k+1)} = (\mathbf{I} - \eta \mathbf{A}_\lambda) \mathbf{e}^{(k)}.$$

Therefore an eigendirection with unpenalized eigenvalue a is multiplied each iteration by

$$1 - \eta(a + \lambda).$$

Convergence requires $|1 - \eta(a + \lambda)| < 1$ for every direction. With $M = 9.09$,

$$0 < \eta < \frac{2}{M} = \frac{2}{9.09} \approx 0.2200.$$

For $\eta = 1/M$, the factors are 0 and

$$1 - \frac{0.10}{9.09} \approx 0.9890,$$

so the high-curvature direction vanishes immediately while the low-curvature direction converges slowly. For $\eta = 2.1/M$, the high-curvature factor is -1.1 , whose magnitude exceeds one, so the iteration diverges in that direction.

Although a larger λ can improve optimization, it also changes the statistical estimator by increasing shrinkage bias. Therefore λ should be chosen for predictive performance, not only for computational speed.

Question 5: Selecting the ridge penalty for real-estate prediction

5.1 Ten-fold cross-validation

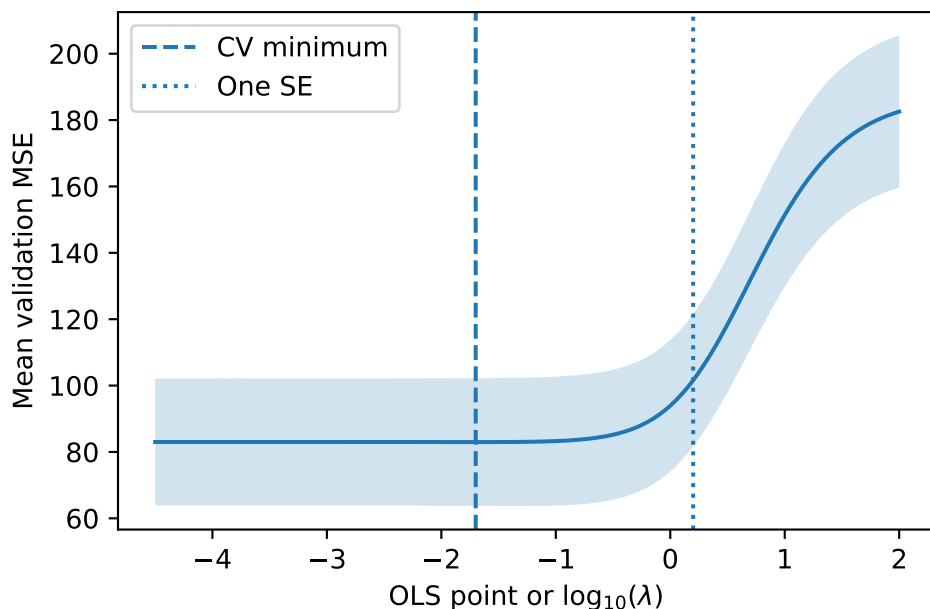
For each validation fold, means and scales are estimated only from the other nine folds, preventing leakage.

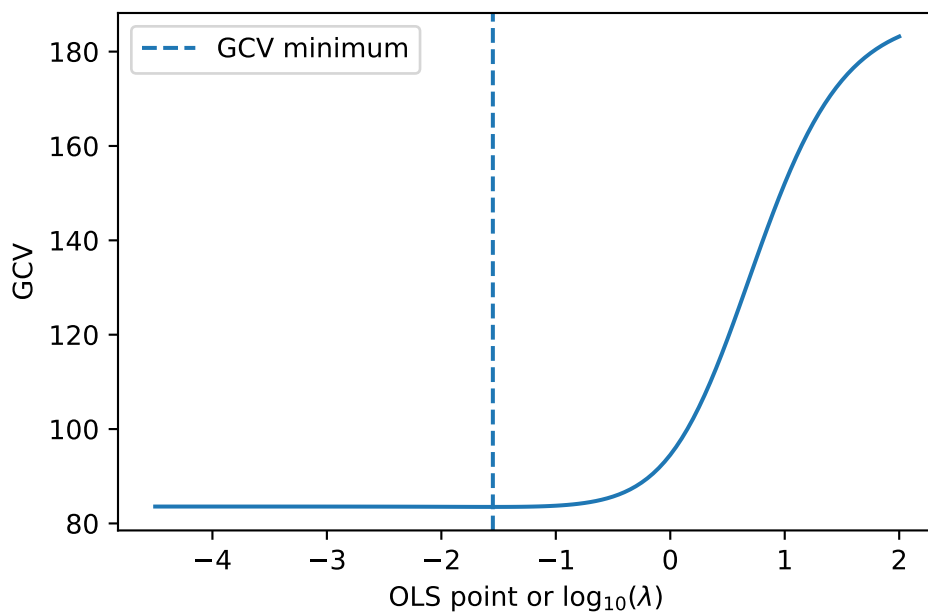
$$\lambda_{\min} = 0.01995, \quad \lambda_{\text{lse}} = 1.58489.$$

5.2 GCV

For GCV, preprocessing is estimated once from all 332 training observations and kept fixed across the penalty path.

$$\lambda_{\text{GCV}} = 0.02818.$$





5.3 Final comparison

The tuning choices were fixed before the test responses were used.

Rule	λ	Effective df	Training MSE	Test MSE
OLS	0	7.000	80.094	66.322
CV minimum	0.01995	6.773	80.135	66.000
One SE	1.58489	3.033	100.374	81.200
GCV	0.02818	6.689	80.171	65.903

OLS has the smallest training MSE. CV-minimum and GCV choose small penalties and have similar test performance, while the one-SE rule regularizes much more strongly. The test MSEs are for final evaluation only: choosing the model after inspecting them would turn the test set into another validation set and bias the reported performance downward.

explain-to-me skill

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name: explain-to-me
description: Explain a homework question to the student without solving it.
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# Explain to Me

When I explicitly ask you to use this skill on a homework question:

1. Explain in simple language what the question is asking me to do.
2. Break the question into smaller tasks.
3. Explain any important notation, terminology, or formulas that appear in the question.

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4. Identify the relevant statistical or mathematical concepts I should review.
5. Suggest a reasonable order for approaching the problem.
6. Point out common misunderstandings or mistakes I should avoid.

Do not solve the homework problem for me.

Do not provide the final numerical answer, derivation, proof, code solution, or completed response.

You may use small illustrative examples that are different from the homework problem if they help.

Stop after explaining what I need to do and what knowledge I need.